europass Curriculum vitae

PERSONAL INFORMATION

Simone Francesco Fornasari

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ORCID 0000-0001-8855-4478

Date of birth 19 January 1994 | Nationality Italian

EDUCATION

2020–2024

PhD in Earth Science, Fluid-Dynamics, and Mathematics. Interactions and Methods

University of Trieste, Department of Mathematics and Geosciences

Cycle XXXVI

Awarded on 22 March 2024

Thesis Development of machine learning approaches for real-time ground-shaking maps reconstruction

Supervisor Giovanni Costa

Co-supervisor Veronica Pazzi

Sector GEO/10 (now GEOS-04/A – Solid Earth Geophysics)

Activities Programming, Machine learning and neural networks, Signal analysis, Real-time ground-shaking maps, Real-time automatic acquisition and analysis of accelerometric data, Seismic noise analysis, Seismic signal denoising

2017–2019

MSc in Geosciences

University of Trieste, Department of Mathematics and Geosciences

Degree class LM-74

Thesis Scenari di scuotimento, stima degli effetti di sito e dell'input sismico in ambito urbano (Ground-shaking scenarios, estimation of site effects and seismic input in urban areas)

Supervisor Fabio Romanelli

Co-supervisor Franco Vaccari

Sector GEO/10 (now GEOS-04/A – Solid Earth Geophysics)

Activities Programming, Numerical simulations

2013–2017

BSc in Physics

University of Trieste, Department of Physics

Degree class L-30

Thesis Analisi e Sintesi di Segnali Sismici Lunari (Analysis and Synthesis of Lunar Seismic Signals)

Supervisor Fabio Romanelli

Co-supervisors Franco Vaccari and Alessandro Vuan

Sector FIS/06 (now PHYS-05/B)

Activities Programming, Signal analysis

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WORK EXPERIENCE

16 November 2023 – present

Postdoctoral Research Fellow (Assegno di Ricerca)

University of Trieste, Department of Mathematics, Informatics and Geosciences

Machine learning techniques applied to the real-time analysis of accelerometric data for civil protection purposes

Sector GEOS-04/A – Solid Earth Geophysics (formerly GEO/10)

Activities Programming, Machine learning and neural networks, Signal analysis, Real-time ground-shaking maps, Earthquake Early Warning and numerical simulations, Spectral decomposition and Bayesian models, Seismic noise analysis, Landslide characterisation

May 2019 – July 2019

Curricular Internship

University of Trieste

Computation of synthetic seismograms for site-response assessment, for engineering applications of seismic-input evaluation. Basic Bash scripting. Understanding of the models used to assess seismic site response.

Sector GEO/10 (now GEOS-04/A – Solid Earth Geophysics)

Activities Programming, Numerical simulations

October 2016 – December 2016

Curricular Internship

National Institute of Oceanography and Applied Geophysics (OGS)

Development of Python codes for the (re-)location of seismic events

Sector FIS/06 (now PHYS-05/B)

Activities Programming

RESEARCH PROJECTS

June 2025 – present

CONCORDIA – *COoperative Network for COoRdinated Disaster Impact Assessment*

Funding Interreg VI-A Italy-Slovenia (2025–2027)

Partner University of Trieste, Department of Mathematics, Informatics and Geosciences (lead partner)

Activities Development of a harmonised ground-shaking map

December 2024 – present

TRANSFORM^2 – *TowaRds AdvaNced multidiSciplinary Fault ObseRvatory systeMs*^{2}

Funding Horizon Europe, HORIZON-INFRA-2024-DEV-01, GA 101188365 (2024–2027)

Partner University of Trieste, Department of Mathematics, Informatics and Geosciences

Activities Feasibility analysis of Earthquake Early Warning for the NITRO Near-Fault Observatory

December 2024 – April 2025

MEET – *Monitoring Earth's Evolution and Tectonics*

Funding Italian National Recovery and Resilience Plan (PNRR), Mission 4, Research Infrastructures, IR0000025 (2022–2025)

Partner University of Trieste, Department of Mathematics, Informatics and Geosciences

Activities Development of the UniTS unit dataset

2021 – present

Collaboration agreements with the Italian Civil Protection Department (DPC), the Friuli Venezia Giulia Civil Protection and EPOS

DPC Accelerometric monitoring in Friuli Venezia Giulia and Veneto and advice on the processing of the Italian Strong-Motion Network (RAN) data (2023–2025; 2025–2027)

PC-FVG Scientific collaboration on environmental and territorial monitoring (2025–2029)

EPOS JRU EPOS Italia agreements with INGV (2021–2026), including services for the North eastern Italy ThRust faults Observatory (NITRO) – NFO-TCS

Partner University of Trieste, Department of Mathematics, Informatics and Geosciences (principal investigator G. Costa)

Activities Participation in the seismic and accelerometric monitoring and data-processing activities covered by the agreements

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TRAINING

September 2026 — *A cinquant'anni dal terremoto del Friuli: quanto siamo preparati a gestire la prossima emergenza sismica?* (Fifty years after the Friuli earthquake: how prepared are we to manage the next seismic emergency?)
Workshop within the INGV-OGS Agreement, INGV, Rome, Italy

June 2025 — *The International Irpinia Summer School: New Monitoring Technologies for Near Fault Observatories (NFO)*
INGV, University of Naples Federico II and University of Bari, within the MEET project, Grottaminarda (AV) and Naples, Italy

November 2024 — *11th Workshop on Collaborative Scientific Software Development and Management of Open Source Scientific Packages*
ICTP, Trieste, Italy

July 2023 — *2023 InSAR Processing and Theory with GMTSAR Short Course*
EarthScope Consortium, virtual

November 2022 — *Public Engagement with Geoscience Workshop Series*
EGU Outreach Committee, virtual

July 2022 — *Machine Learning in Geosciences Summer School*
University of Pisa, Pisa, Italy

July 2022 — *Eastern European Machine Learning 2022 Summer School*
EEML22, Vilnius, Lithuania (virtual)

October 2021 — *Data Assimilation and Inverse Problems in Geophysical Sciences Workshop*
Joint ICTP-IUGG (virtual)

PUBLICATIONS

A26b — Veronica Pazzi, **Simone Francesco Fornasari**, Stefano Devoto, Giovanni Costa, and Emanuele Forte. "Reply to: On the Use of Seismic Noise Studies for Landslide Rock Block Hazard Assessment". In: *Communications Earth & Environment* 7.1 (Sept. 18, 2026), p. 767. DOI: 10.1038/s43247-026-04028-3

A26a — **Simone Francesco Fornasari** and Giovanni Costa. "Assessing Earthquake Early Warning Feasibility at the NITRO Near-Fault Observatory Using Numerical and Offline Simulations". In: *Seismological Research Letters* (July 2026). DOI: 10.1785/0220260091

A25c — Veronica Pazzi, **Simone Francesco Fornasari**, Stefano Devoto, Giovanni Costa, and Emanuele Forte. "Fast Estimation of Landslide Blocks' Volume from Seismic Noise Measurements". In: *Communications Earth & Environment* 6.1 (Dec. 2025), p. 1031. DOI: 10.1038/s43247-025-02999-3

A25b — Deniz Ertuncay, Pinar Buyukakpinar, **Simone Francesco Fornasari**, and Onur Tan. "23 April 2025 Marmara Sea (Mw 6.3), Türkiye Earthquake: Mainshock, Aftershock, and Ground Observations". In: *Seismica* 4.2 (Nov. 2025). DOI: 10.26443/seismica.v4i2.1773

A25a — Deniz Ertuncay, **Simone Francesco Fornasari**, and Giovanni Costa. "Effect of the COVID-19 Lockdown on Background Noise Levels in Italian Strong Motion Network". In: *Frontiers in Earth Science* 12 (Jan. 2025), p. 1507241. DOI: 10.3389/feart.2024.1507241

A24a — **Simone Francesco Fornasari**, Veronica Pazzi, and Giovanni Costa. "Neural-Network and Multivariate-Normal-Distribution Hybrid Method for Real-Time Ground-Shaking Reconstruction". In: *Bulletin of the Seismological Society of America* 114.6 (Sept. 2024), pp. 2912–2925. DOI: 10.1785/0120240095

A23a — **Simone Francesco Fornasari**, Deniz Ertuncay, and Giovanni Costa. "Seismic background noise levels in the Italian strong-motion network". In: *Natural Hazards and Earth System Sciences* 23.10 (2023), pp. 3219–3234. DOI: 10.5194/nhess-23-3219-2023

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A22b **Simone Francesco Fornasari**, Veronica Pazzi, and Giovanni Costa. "A Machine-Learning Approach for the Reconstruction of Ground-Shaking Fields in Real Time". In: *Bulletin of the Seismological Society of America* 112.5 (July 2022), pp. 2642–2652. DOI: 10.1785/0120220034

A22a Giovanni Costa, Piero Brondi, Laura Cataldi, Stefano Cirilli, Arianna Cuius, Deniz Ertuncay, Piero Falconer, Luisa Filippi, **Simone Francesco Fornasari**, Veronica Pazzi, and Philippe Turpaud. "Near-Real-Time Strong Motion Acquisition at National Scale and Automatic Analysis". In: *Sensors* 22.15, 5699 (2022). DOI: 10.3390/s22155699

CONFERENCES

Abstracts

B26a **Simone Francesco Fornasari** and Giovanni Costa. "A Feasibility Study of Earthquake Early Warning at the NITRO Near-Fault Observatory". EGU General Assembly 2026, Vienna, Austria, EGU26-12472. 2026. DOI: 10.5194/egusphere-egu26-12472

B25a **Simone Francesco Fornasari** and Giovanni Costa. "Site response function evaluation at Campi Flegrei accelerometric stations". EGU General Assembly 2025, Vienna, Austria, EGU25-11427. 2025. DOI: 10.5194/egusphere-egu25-11427

B23c **Simone Francesco Fornasari**, Veronica Pazzi, and Giovanni Costa. "Development of an hybrid GMPE-less ShakeMap implementation for real-time ground shaking maps reconstruction". EGU General Assembly 2023, Vienna, Austria, EGU23-15719. 2023. DOI: 10.5194/egusphere-egu23-15719

B23b **Simone Francesco Fornasari**, Veronica Pazzi, and Giovanni Costa. "Real-Time Ground Shaking Maps Reconstructions With a Hybrid ShakeMap Implementation". SSA Annual Meeting 2023, San Juan, Puerto Rico. Seismological Research Letters, 94(2B), 1258. 2023. DOI: 10.1785/0220230054

B22a **Simone Francesco Fornasari**, Giovanni Costa, and Veronica Pazzi. "A machine-learning approach for the reconstruction of ground shaking fields in real-time". EGU General Assembly 2022, Vienna, Austria, EGU22-2673. 2022. DOI: 10.5194/egusphere-egu22-2673

Co-authored abstracts

B26c Alessia Scalabrini, **Simone Francesco Fornasari**, and Giovanni Costa. "Cormons landslide characterization using Lidar and remote sensed data". EGU General Assembly 2026, Vienna, Austria, EGU26-6955. 2026. DOI: 10.5194/egusphere-egu26-6955

B26b Veronica Pazzi, **Simone Francesco Fornasari**, Stefano Devoto, Giovanni Costa, and Emanuele Forte. "Rapid estimation of block volumes from seismic noise measurements and an eigenfrequency abacus". EGU General Assembly 2026, Vienna, Austria, EGU26-17252. 2026. DOI: 10.5194/egusphere-egu26-17252

B25c Federico Parentelli, Chantal Beltrame, **Simone Francesco Fornasari**, Veronica Pazzi, Giorgia Moschion, and Giovanni Costa. "Urban seismic network development for site effects evaluation in Trieste". EGU General Assembly 2025, Vienna, Austria, EGU25-19491. 2025. DOI: 10.5194/egusphere-egu25-19491

B25b Seyedmohammadsadegh Jafari, Deniz Ertuncay, **Simone Francesco Fornasari**, Laura Cataldi, Veronica Pazzi, and Giovanni Costa. "Empirical Attenuation Characteristics and Seismic Source Parameters through Spectral Inversion for Northeastern Italy". EGU General Assembly 2025, Vienna, Austria, EGU25-17578. 2025. DOI: 10.5194/egusphere-egu25-17578

B24a Federico Parentelli, Philippe Turpaud, **Simone Francesco Fornasari**, Deniz Ertuncay, Arianna Cuius, Giovanni Costa, and Veronica Pazzi. "Seismic noise measurements in the city of Trieste in order to plan an urban seismic network". EGU General Assembly 2024, Vienna, Austria, EGU24-7933. 2024. DOI: 10.5194/egusphere-egu24-7933

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B23d Deniz Ertuncay, **Simone Francesco Fornasari**, and Giovanni Costa. "Seismic Background Noise of Italian Strong Motion Network". SSA Annual Meeting 2023, San Juan, Puerto Rico. Seismological Research Letters, 94(2B), 1227. 2023. DOI: 10.1785/0220230054

B23a Deniz Ertuncay, **Simone Francesco Fornasari**, and Giovanni Costa. "Seismic Signal Denoising with Attention U-Net". EGU General Assembly 2023, Vienna, Austria, EGU23-1309. 2023. DOI: 10.5194/egusphere-egu23-1309

Conference proceedings

C26a **Simone Francesco Fornasari** and Giovanni Costa. "Assessing Earthquake Early Warning Feasibility at the NITRO Near-Fault Observatory". In: *GNGTS-BOOK OF ABSTRACTS 44th NATIONAL CONFERENCE*. 2026

C25a **Simone Francesco Fornasari** and Giovanni Costa. "Estimation of the site response function for accelerometric stations within Campi Flegrei". In: *GNGTS-BOOK OF ABSTRACTS 43rd NATIONAL CONFERENCE*. 2025

C24a **Simone Francesco Fornasari**, Veronica Pazzi, and Giovanni Costa. "Development of a hybrid method for ground shaking map reconstruction in near-real time". In: *GNGTS-BOOK OF ABSTRACTS 42nd NATIONAL CONFERENCE*. Alessandro Rebez & Giulia Massolino. 2024, pp. 578–584

C23a **Simone Francesco Fornasari**, Veronica Pazzi, and Giovanni Costa. "Real-time ground-shaking maps reconstruction of late 2022 Italian earthquakes". In: *GNGTS-BOOK OF ABSTRACTS 41st NATIONAL CONFERENCE*. Alessandro Rebez & Giulia Massolino. 2023, pp. 768–773

Co-authored conference proceedings

C26c Federico Parentelli, Chantal Beltrame, **Simone Francesco Fornasari**, and Giovanni Costa. "Site effects characterisation for the city of Trieste". In: *GNGTS-BOOK OF ABSTRACTS 44th NATIONAL CONFERENCE*. 2026

C26b Veronica Pazzi, **Simone Francesco Fornasari**, Stefano Devoto, Giovanni Costa, and Emanuele Forte. "An eigenfrequency abacus for rapid estimation of block volumes from seismic noise measurements". In: *GNGTS-BOOK OF ABSTRACTS 44th NATIONAL CONFERENCE*. 2026

C25b Federico Parentelli, Chantal Beltrame, **Simone Francesco Fornasari**, Veronica Pazzi, Giorgia Moschion, and Giovanni Costa. "A new urban seismic network for the city of Trieste. Where to place it?" In: *GNGTS-BOOK OF ABSTRACTS 43rd NATIONAL CONFERENCE*. 2025

Invited talks

D24a **Simone Francesco Fornasari**, Veronica Pazzi, and Giovanni Costa. "Real-time Ground Shaking Maps With Antelope And ML". Oral presentation at European Antelope Users Group Meeting 2024, Vienna, Austria. 2024

D23a **Simone Francesco Fornasari**, Veronica Pazzi, and Giovanni Costa. "The road to real time ground shaking maps is paved with ORB packages". Oral presentation at European Antelope Users Group Meeting 2023, Vienna, Austria. 2023

Oral presentations

E26e **Simone Francesco Fornasari** and Giovanni Costa. "Investigating the Potential for Earthquake Early Warning at the NITRO Near-Fault Observatory". Oral presentation at 40th General Assembly of the European Seismological Commission, Istanbul, Türkiye. 2026

E26d **Simone Francesco Fornasari** and Giovanni Costa. "Bayesian Generalised Linear Models for Spectral Decomposition at Campi Flegrei". Oral presentation at 40th General Assembly of the European Seismological Commission, Istanbul, Türkiye. 2026

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E26c **Simone Francesco Fornasari** and Giovanni Costa. "Evaluating NITRO's EEWS readiness". Short oral presentation (2–3 min) at EPOS Days 2026, Cagliari, Italy. 2026

E26b Veronica Pazzi, **Simone Francesco Fornasari**, Stefano Devoto, Giovanni Costa, and Emanuele Forte. "An eigenfrequency abacus for rapid estimation of block volumes from seismic noise measurements". Oral presentation (speaker: S. F. Fornasari) at GNGTS26, Udine, Italy. 2026

E26a **Simone Francesco Fornasari** and Giovanni Costa. "Assessing Earthquake Early Warning Feasibility at the NITRO Near-Fault Observatory". Oral presentation at GNGTS26 (and AGLC26), Udine, Italy. 2026

E25b **Simone Francesco Fornasari** and Giovanni Costa. "Site response function evaluation at Campi Flegrei accelerometric stations". Oral presentation at EGU General Assembly 2025, Vienna, Austria. 2025

E25a **Simone Francesco Fornasari** and Giovanni Costa. "Estimation of the site response function for accelerometric stations within Campi Flegrei". Oral presentation at GNGTS25 (and AGLC25), Bologna, Italy. 2025

E24a **Simone Francesco Fornasari**, Veronica Pazzi, and Giovanni Costa. "Development of a hybrid method for ground shaking map reconstruction in near-real time". Oral presentation at GNGTS24 (and AGLC24), Ferrara, Italy. 2024

E23c **Simone Francesco Fornasari**, Veronica Pazzi, and Giovanni Costa. "Development of a hybrid GMPE-less ShakeMap implementation for real-time ground shaking maps reconstruction". Oral presentation at EGU General Assembly 2023, Vienna, Austria. 2023

E23b **Simone Francesco Fornasari**, Veronica Pazzi, and Giovanni Costa. "Real-Time Ground Shaking Maps Reconstructions With a Hybrid ShakeMap Implementation". Oral presentation at SSA Annual Meeting 2023, San Juan, Puerto Rico. 2023

E23a **Simone Francesco Fornasari**, Veronica Pazzi, and Giovanni Costa. "Real-time ground-shaking maps reconstruction of late 2022 Italian earthquakes". Oral presentation at GNGTS23 (and AGLC23), Bologna, Italy. 2023

E22a **Simone Francesco Fornasari**, Giovanni Costa, and Veronica Pazzi. "A machine-learning approach for the reconstruction of ground shaking fields in real-time". Oral presentation at EGU General Assembly 2022, Vienna, Austria (virtual). 2022

Posters

F26b **Simone Francesco Fornasari** and Giovanni Costa. "A Feasibility Study of Earthquake Early Warning at the NITRO Near-Fault Observatory". Poster at EGU General Assembly 2026, Vienna, Austria. 2026

F26a **Simone Francesco Fornasari** and Giovanni Costa. "Evaluating NITRO's EEWS readiness". Poster at EPOS Days 2026, Cagliari, Italy. 2026

F23a **Simone Francesco Fornasari**, Veronica Pazzi, and Giovanni Costa. "Toward a general hybrid ground shaking map reconstruction method". Poster at BeGEO 2023, Naples, Italy. 2023

F22a **Simone Francesco Fornasari**. "Investigating machine-learning based mapping techniques for seismic data denoising". Poster (in collaboration with Deniz Ertuncay) at EEML 2022, Vilnius, Lithuania (virtual). 2022

HONOURS AND AWARDS

2024 Special mention, *Premio Associazione per la Geofisica Licio Cernobori (AGLC) 2024*
GNGTS 2024, Ferrara, Italy

2022 *Outstanding Student and PhD candidate Presentation (OSPP) Award 2022*
EGU General Assembly 2022, Vienna, Austria

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PEER REVIEW

Reviewer for *Geophysical Journal International*, *Natural Hazards and Earth System Sciences* and *International Journal of Geophysics*

SESSION CONVENING

May 2026 — EGU General Assembly 2026
Session SM8.3 – "Assessment of Earthquake Related Hazards, Site Effects, and Microzonation"
Vienna, Austria

May 2025 — EGU General Assembly 2025
Session SM8.1 – "Assessment of Earthquake Related Hazards, Site Effects, and Microzonation"
Vienna, Austria

April 2024 — EGU General Assembly 2024 (co-convener)
Session SM8.1 – "Assessment of Earthquake Related Hazards, Site Effects, and Microzonation"
Vienna, Austria

October 2023 — BeGEO 2023
Session "Seismology"
University of Naples Federico II, Naples, Italy

OUTREACH

November 2022 – present — Secondary-school workshop "Come nascono i terremoti e come affrontare il rischio sismico" (How earthquakes originate and how to deal with seismic risk)
Trieste, Italy

September 2026 — Science festival *Trieste Next*
Stand "50 anni dopo il terremoto del Friuli. Il contributo dell'Università di Trieste alla conoscenza dei terremoti" (50 years after the Friuli earthquake: the University of Trieste's contribution to the knowledge of earthquakes)
Trieste, Italy

September 2023 — Science festival *Trieste Next*
Stand "Geoscienze e sostenibilità ambientale" (Geosciences and environmental sustainability)
Trieste, Italy

September – October 2022 — Exhibition *Terremoti d'Italia* (Earthquakes of Italy)
During the *Trieste Next* science festival and Barcolana 2022
Trieste, Italy

September 2021 — Science festival *Trieste Next*
Stand "Sicurezza geologica, terremoti, reti di monitoraggio sismico e mappe di scuotimento" (Geological safety, earthquakes, seismic monitoring networks and ground-shaking maps)
Trieste, Italy

PERSONAL SKILLS

Mother tongue: Italian

Other languages

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken interaction	Spoken production	
English	C1	C1	B2	B2	C1

Levels: A1 and A2: Basic user – B1 and B2: Independent user – C1 and C2: Proficient user
Common European Framework of Reference for Languages

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Digital competences

SELF-ASSESSMENT				
Information Processing	Communication	Content creation	Safety	Problem solving
Advanced user	Advanced user	Advanced user	Advanced user	Advanced user

Digital competences - Self-assessment grid

Computer skills

- Python for scientific computing, seismology and machine learning
- Bayesian modelling (PyMC)
- BRTT Antelope: data acquisition and management, and development of integrated codes for real-time analysis
- USGS ShakeMap and Earthquake Early Warning software (PRESTo)
- Standard formats and services for seismological data (miniSEED, StationXML, FDSN web services)
- Collaborative source-code management (Git, GitHub, GitLab)
- Unix/Linux and Windows operating systems; Bash scripting; cluster computing (SLURM)
- Containerisation (Docker)
- GIS and cartography (QGIS)
- LaTeX
- Vector graphics and 3D modelling software

Driving licence A1, B

I authorise the processing of the personal data contained in my curriculum vitae in accordance with Regulation (EU) 2016/679 (GDPR).

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